


46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

**Sec: SR STAR - A & APEX, ALL IITS** **DT: 23/03/2026**
**Time: 3 Hrs** **GTM - 02** **Max. Marks: 300**

**PHYSICS** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

1. Three equations A, B and C are measured as $A = 2 \pm 0.005, B = 1 \pm 0.001$ and $C = 4 \pm 0.01$. Then the percentage error in $R = \frac{AC^3}{B^2}$ is

A) 1.2 B) 0.8 C) 0.7 D) 1.05

2. In Young's double slit experiment how many maximas can be obtained on a screen (including the central maximum) on both sides of the central fringe if λ = 2000 Å and d = 7000 Å

A) 12 B) 7 C) 18 D) 4

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3. Two identical capacitors A and B are connected in series with an ideal battery. When separation between plates of capacitor A is doubled then choose incorrect option(s).

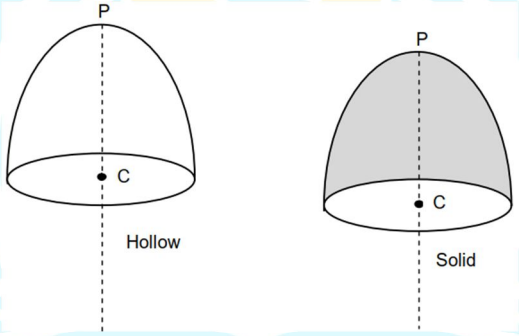
(A) Battery will act as load in this process.

(B) Potential difference between plates of capacitor A increases.

(C) Potential difference between plates of capacitor B decreases.

(D) Energy of capacitor 'A' decreases

4. Consider uniformly positively charged hollow hemisphere and solid hemisphere of same radius.




P is a point on circumference and C is center then choose incorrect option.

(A) in case of hollow sphere there is no point between C and P on line CP where Electric field is zero.

(B) in case of solid sphere there is a point between C and P on line CP where Electric field is zero.

(C) when we move from C to P in hollow hemisphere electric potential continuously increases

(D) when we move from C to P in solid hemisphere electric potential continuously increases

5. A steel ball initially at a pressure of $10^5$ Pa is heated from $20\,^0C$ to $120\,^0C$ keeping its volume constant. Find the final pressure inside the ball. Given that coefficient of linear expansion of steel is $1.1\times10^{-5}\ {}^0C^{-1}$ and Bulk modulus of steel is $1.6\times10^{11}\ Nm^{-2}$.

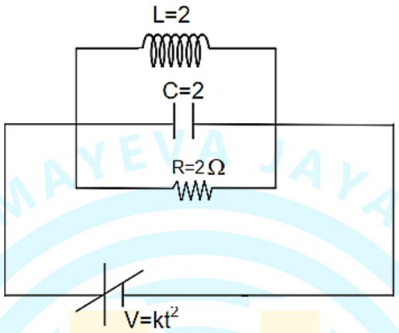
A) $5.28\times10^8\ Pa$ B) $5.28\times10^8\ Pa$

C) $2.64\times10^5\ Pa$ D) $2.64\times10^6\ Pa$



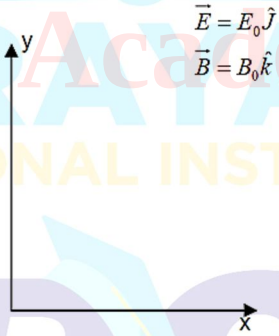
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6. Source of variable voltage starts t = 0. initially no energy in capacitor and inductor. Find current through battery at t = 2 sec. $\left(k = 1\frac{volt}{sec^2}\right)$




(A) $\frac{34}{3}$ amp (B) $\frac{17}{3}$ amp (C) $\frac{2}{3}$ amp (D) 2 amp

7. A unit positive charge of mass 'm' is released at origin. There is a transverse uniform electromagnetic field as shown. When charged particle has its maximum y – coordinate is angular momentum about origin is L. Then its maximum y – coordinate is

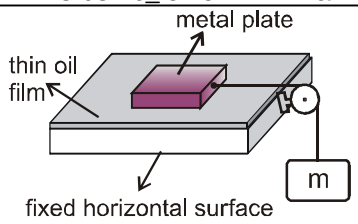



(A) $\left(\frac{L^2}{mE}\right)^{\frac{4}{3}}$ (B) $\left(\frac{L^2}{mE}\right)^{\frac{1}{3}}$ (C) $\left(\frac{L^2}{mE}\right)^{\frac{2}{3}}$ (D) $\left(\frac{L^2}{2mE}\right)^{\frac{1}{3}}$

8. A rectangular metal plate has dimensions of 10 cm × 20 cm. A thin film of oil separates the plate from a fixed horizontal surface. The separation between the rectangular plate and the horizontal surface is 0.2 mm. An ideal string is attached to the plate and passes over an ideal pulley to a mass m. When m = 125 gm, the metal plate moves at constant speed of 5 cm/s across the horizontal surface. Then the coefficient of viscosity of oil in dyne–s/cm$^2$ is (Use g=1000 cm/s$^2$)



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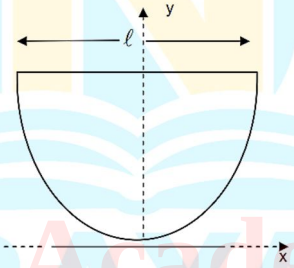



(A) 5 (B) 25 (C) 2.5 (D) 50

9. The ratio of minimum to maximum wavelength in Balmer series is

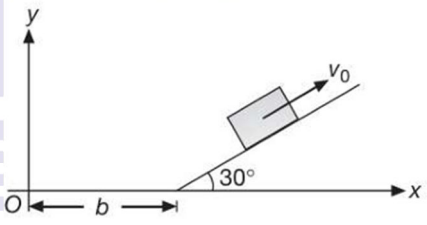
A) $\frac{5}{9}$ B) $\frac{5}{36}$ C) $\frac{1}{4}$ D) $\frac{3}{4}$

10. An infinite length wire is lying along x – axis and a loop whose curved part is parabolic of type $y = x^2$. vertex of parabola is just at origin as shown in figure. Mutual inductance of system is.




(A) $\frac{2\mu_0}{\pi}\cdot l$ (B) $\frac{\mu_0}{\pi}\cdot l$ (C) $\frac{\mu_0}{2\pi}\cdot l$ (D) $\frac{4\mu_0}{\pi}\cdot l$

11. A cube of mass m and side a is moving along a plane with constant speed $v_0$ as shown in figure. The magnitude of angular momentum of the cube about z-axis would be



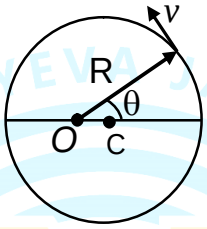

A) $\frac{mv_0b}{2}$ B) $\frac{\sqrt{3}mv_0b}{2}$

C) $mv_0\left(b-\frac{a}{2}\right)$ D) $\frac{mv_0(b-a)}{2}$



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12. A particle is revolving in a circular path as shown in figure in the horizontal plane such that angular velocity of the particle about the point O is constant and is equal to 1 rad/s. Distance of the particle from O is given by $R = R_0 + \beta t$ where $R_0$ and are constant. The speed of the particle as a function of time is :




(A) $\sqrt{\beta^2 + 1}$ (B) $\left(R_0 + \beta t\right)$

(C) $\sqrt{\beta^2 + \left(R_0 + \beta t\right)^2}$ (D) $\beta$

13. Electromagnetic waves with wavelength

(i) $\lambda_1$ is used in satellite communication.

(ii) $\lambda_2$ is used in LASIK eye surgery.

(iii) $\lambda_3$ is used to detect leakage of oil in underground pipelines.

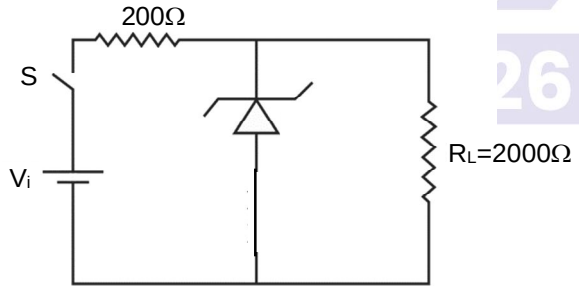
(iv) $\lambda_4$ is used to improve visibility in optical communication.

Then arrange these wavelengths in ascending order of their magnitude.

(A) $\lambda_3 < \lambda_2 < \lambda_4 < \lambda 1$ (B) $\lambda_2 < \lambda_1 < \lambda_3 < \lambda_4$

(C) $\lambda_1 < \lambda_4 < \lambda_3 < \lambda_2$ (D) $\lambda_4 < \lambda_1 < \lambda_2 < \lambda_3$

14. The switch S is closed. Over what range of input voltage will the zener circuit shown in figure maintain 30V across the load $R_L$, assuming that series resistance R = 200Ω and zener current rating is 25mA?




(A) (22 – 28) V (B) (25 – 30) V (C) (33 – 38) V (D) (36 – 43) V



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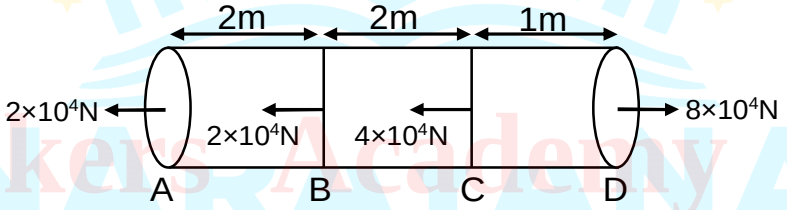
15. Internal resistance of battery of emf ε is given by r = k i where k is positive constant and i is current drawn from battery. Maximum output power withdrawn from battery by a load resistor is –

(A) $\frac{\varepsilon}{3}\sqrt{\frac{\varepsilon}{3k}}$ (B) $\frac{2\varepsilon}{3}\sqrt{\frac{\varepsilon}{3k}}$ (C) $\frac{\varepsilon}{3}\sqrt{\frac{2\varepsilon}{3k}}$ (D) $\frac{2\varepsilon}{3}\sqrt{\frac{\varepsilon}{k}}$

16. A neutron moving with kinetic energy k collides head on with $He^+$ atom which is initially at rest, and in ground state. If neutron rebounds back with one fourth of its kinetic energy after collision then minimum value of k so that two photons can be obtained in collision is approximately

(A) 65 eV (B) 258 eV (C) 165 eV (D) 60 eV

17. If uniform rod of length 5m, area of cross-section $50cm^2$ and young's modulus $4 \times 10^9$ $N/m^2$ is lying at rest under the action of forces as shown in figure, then total extension in the rod is.




(A) 2 cm (B) 0.5 cm (C) 1.5 cm (D) 1 cm

18. Consider the following statements about photodiodes, solar cells and LEDs:

a. Photodiodes are normally operated in reverse bias.

b. The depletion region in a solar cells is made very wide to absorb more light.

c. LEDs are heavily doped p-n junctions.

d. In photodiodes, the photocurrent increases almost linearly with incident light intensity even at high illumination

e. Solar cells can work without any external bias.

Which of the following options contain only correct statements?

A) a, b, e B) a, c, e C) a, d, e D) b, c, d



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19. An equiconvex lens, having radius of curvature 33 cm, is placed on a horizontal plane mirror and a pin held 20 cm above the lens coincides with its image. Now the space between the lens and the mirror is filled with a liquid. In order to coincide with the image the pin has to be raised by 5 cm. The refractive index of the liquid is

A) 1.33 B) 1.53 C) 2.33 D) 2.66

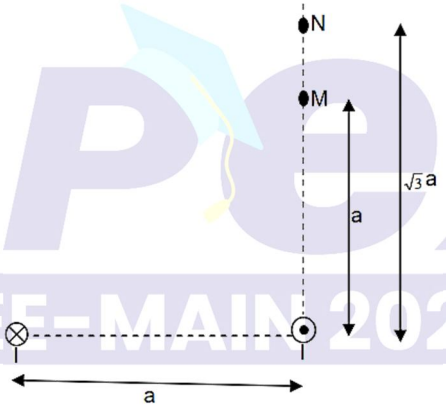
20. The total heat supplied and the change in internal energy of a monoatomic gas during an isobaric process are $E_1$ and $E_2$ respectively. If $\frac{E_1}{E_2}=\frac{x}{6}$ then the value of x is___

A) 10 B) 5 C) 15 D) 20

## SECTION – II
## (Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

21. Two parallel infinite wires carrying same current in opposite direction placed perpendicular to plane of paper. There are two points on a line perpendicular to line joining wire as shown in figure. angle between direction of magnetic field at M and N is _____ degree.



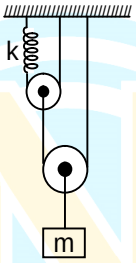

22. If gravitation potential enemy of a uniform cube of side 'a' is U then work done to bring such 8 cubes form infinite separation to form another cube of side '2a' is given by kU where k = _____



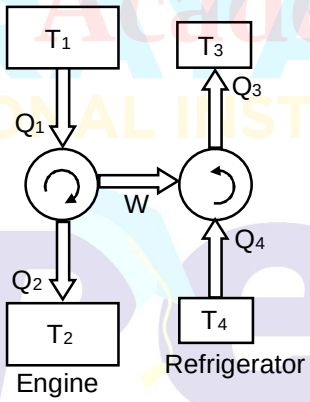
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23. A closely wound coil has a radius of 6.00 cm and carries a current of 2.50 A. How many turns must it have if, at a point on the coil axis 6.00 cm from the centre of the coil, the magnetic field is $6.39\times10^{-4}T$ ?

24. The natural frequency of oscillation of the system as shown in figure is $\frac{x}{\pi}\sqrt{\frac{k}{m}}$ where x is an integer. Pulleys are massless and frictionless. Spring and string are also massless. Then the value of 8x is




25. Figure represents a Carnot engine that works between temperatures $T_1$ = 400 K and $T_2$ = 150 K and drives a Carnot refrigerator that works between temperatures $T_3$ = 320 K and $T_4$ = 300 K. What is the ratio $10\left(\frac{Q_3}{Q_1}\right)$?






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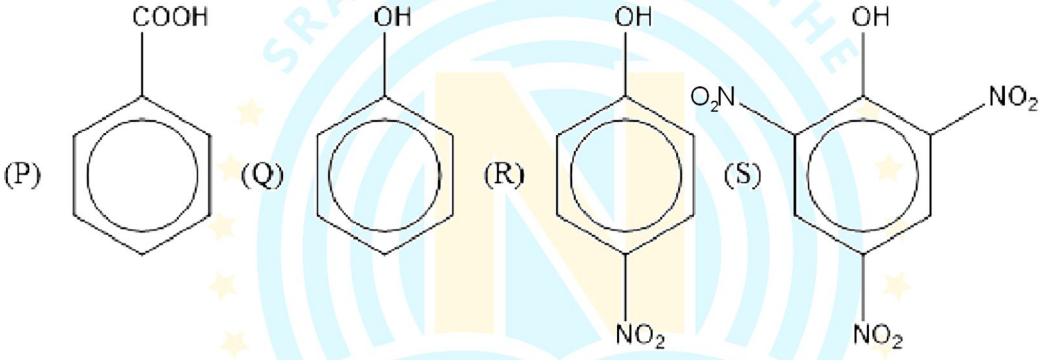
**CHEMISTRY** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

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26. The correct order of pKa values for following is




(A) S > P > R > Q (B) P > S > R > Q

(C) P < S < R < Q (D) S < P < R < Q

27. For decomposition of a gas, the data for half life and initial pressure is -

| $P_0$ ( torr ) → | 50 | 100 | 200 | 400 |
|---|---|---|---|---|
| Relative $t_{1/2}(s)$ → | 40 | 20 | 10 | 5 |

The order of reaction is

(A) 0 (B) 1 (C) 2 (D) 3

28. Energy of $He^{\oplus}$ ion in an excited state where the path length is twice the debroglie's wavelength, is

(A) –3.4 eV (B) –54.4 eV (C) –13.6 eV (D) –27.2 eV

29. Which of the following has –ve electron gain enthalpy?

(A) Be (B) Mg (C) N (D) P

30. Which of the following molecule is a zwitterion?

(A) Benzanilide

(B) Glycine

(C) Fructose

(D) Tetramethyl ammonium chloride



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31. Iodoform test is not given by

(A) Acetone (B) Ethanal (C) Ethanol (D) Ethanoic acid

32. Which of the following is an acidic reagent?

(A) Bayer's Reagent (B) Fehling's solution

(C) Lucas Reagent (D) Tollen's Reagent

33. The complex $\left[Co(NH_3)_3(NO_2)_3\right]$ can show -

(A) Only structural isomerism

(B) Only stereoisomerism

(C) Both structural and stereoisomerism

(D) Neither

34. $CO_2$ gas and $SO_2$ gas cannot be differentiated by

(A) Odour (B) $KMnO_4$ solution

(C) $K_2Cr_2O_7$ solution (D) Lime water

35. $(CH_3)_2C=CH_2 \xrightarrow{KMnO_4/H^{\oplus}}$ Products

Which is not obtained in the products

(A) A gas which turns lime water milky.

(B) A liquid which turns anhydrous $CuSO_4$ blue

(C) A liquid which gives yellow ppt with $I_2$ + NaOH.

(D) A gas which gives red ppt of $Cu_2O$ with Fehling's solution

36. Which option contains both non-planar shapes?

(A) $NO_3^{\ominus}, H_2O_2$ (B) $BF_3$, $PCl_3$ (C) $XeF_4$, $SF_4$ (D) $XeO_3$, $SO_3^{2-}$

37. Assertion (A) : Ethers cannot be distilled upto dryness due to fear of explosion.

Reason (R) : Due to formation of superoxide, ether is explosive.

(A) Both (A) & (R) are correct & (R) is the correct explanation of (A)

(B) Both (A) & (R) are correct & (R) is not the correct explanation of (A)

(C) (A) is correct but (R) is incorrect

(D) (R) is correct but (A) is incorrect



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38. During analysis of salt with cation $M^{2+}$, addition of a reagent (X) to alkaline solution of the salt gives a bright rosy red precipitate. Reagent (X) and $M^{2+}$ are

(A) Dimethyl glyoxime and $Ni^{2+}$ (B) Dimethyl glyoxime and $Co^{2+}$

(C) Nesseler's reagent and $Hg^{2+}$ (D) Nesseler's reagent and $Ni^{2+}$

39. $(P)[C_4H_9Cl_2] \xrightarrow[100°C]{\text{Hydrolysis}} (Q)[C_4H_8O]$

Q reacts with hydroxyl amine but does not give Ag -mirror test with Tollen's reagent. Then (P) & (Q) are respectively

(A) 1,1–Dichlorobutane & Butanal (B) 2,2–Dichlorobutane & Butanal

(C) 1,1–Dichlorobutane & Butan-2-one

(D) 2,2–Dichlorobutane & Butan-2-one

40. When a hydrocarbon undergoes combustion in the presence of air, it requires 9.5 equivalents of oxygen and produces 3 equivalents of water. What is its molecular formula?

(A) $C_8H_6$ (B) $C_9H_9$ (c) $C_6H_6$ (D) $C_9H_6$

41. Which one of the following compounds can show optical isomerism

(A) $CH_3 - CH = C = CH_2$ (B) $CH_2 = CH - CH = CH_2$

(C) $Cl - CH = C = CH - Cl$ (D) $Cl - CH = C = CCl_2$

42. Select the correct statement among the following

A) $O_2F_2$ is thermally more stable than $OF_2$

B) $I_2O_5$ is used in the estimation of carbon dioxide

C) Chemical formula of tear gas is $CCl_3NO_2$

D) Boiling point of HCl is more than HF

43. 1 mole of an ideal gas is expanded from an initial pressure of 1 bar to final pressure of 0.1 bar at constant temperature of $0°C$. Identify the incorrect one for the process

(A) $\Delta U = 0$ (B) $\Delta S = 0$ (C) $\Delta H = 0$ (D) $\Delta(PV) = 0$



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44. Which is not obtained on heating $K_2Cr_2O_7(s)$

(A) Red Salt

(B) Paramagnetic gas

(C) A compound with +6 ON of Cr

(D) A compound with +3 ON of Cr

45. Which of the following increases with increase in temperature

(A) Solubility of gas in water

(B) Resistance of aqueous NaCl solution

(C) Ionic product of water (D) Viscosity of a liquid

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46. At 25°C, find the pH of a solution obtained by mixing 100 mL of 0.1 M $(NH_4)_2SO_4$ and 200 mL of 0.05 M $NH_3$ (aq.). Given $pK_b(NH_3) = 4.7, \log 2 = 0.3$

47. Calculate the total number of bonds and lone pair of electrons in $XeOF_4$

48. Two miscible liquids A and B have vapour pressures of 50 and 100 torr respectively. If A has 0.4 mole fraction in liquid phase then what is the mole percent of B in vapour phase.

49. Oxidation state of central Bromine in tribromo octoxide $(Br_3O_8)$ is____

50. Given (i) $M^{2+} + 2e^- \longrightarrow M(s), E^0 = 0.83\ V,$

(ii) $MCl_4^{2-} + 2e^- \longrightarrow M(s) + 4Cl^{\ominus}, E^{\circ} = 0.65\ V,$

If $\frac{2.303RT}{F} = 0.06\ V$ and,

(iii) $M^{2+} + 4Cl^{\ominus} \rightleftharpoons [MCl_4]^{2-}; K_{eq} = 10^x$

Calculate the value of x.



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**MATHEMATICS** **MAX.MARKS: 100**

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51. The distance of the point (7, –2, 11) from the line $\frac{x-6}{1}=\frac{y-4}{0}=\frac{z-8}{3}$ along the line $\frac{x-5}{2}=\frac{y-1}{-3}=\frac{z-5}{6}$, is:

(A) 12 (B) 14 (C) 18 (D) 21

52. The value of $\int_0^3 \left|x^2-3x+2\right| dx$ is

(A) $\frac{1}{6}$ (B) $\frac{11}{3}$ (C) $\frac{11}{6}$ (D) None

53. The minimum number of elements that must be added to the relation R = {(a, b), (b, c), (b, d)} on the set {a, b, c, d} so that it is an equivalence relation, is

(A) 10 (B) 9 (C) 8 (D) 13

54. The equation $(5x-1)^2+(5y-2)^2=\left(\lambda^2-4\lambda+4\right)(3x+4y-1)^2$ represents an ellipse, if $\lambda \in$

(A) (0, 1] (B) (–2, –1) (C) (2, 3) (D) (–1, 0)

55. $\int\left(\frac{x^2-x+1}{x^2+1}\right)e^{\cot^{-1}x}dx=f(x)e^{\cot^{-1}x}+c$ where c is constant of integration, then f(x) is equal to

(A) –x (B) $\sqrt{1-x}$ (C) x (D) $\sqrt{1+x}$

56. If $f(x)=\max\left\{\sin x, \sin^{-1}(\cos x)\right\}$, then :

(A) f is differentiable everywhere

(B) f is continuous every where but not differentiable

(C) f is discontinuous at $x=\frac{n\pi}{2}, n\in I$

(D) f is non-differentiable at $x=\frac{n\pi}{2}, n\in I$



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57. The value of $\lim\limits_{x\to 0^+}\left(\frac{\cot^{-1}\left(\ln\{x\}\right)}{\pi}\right)^{-\ell n\{x\}}$ (Where {.} denotes fractional part function) is equal to :

(A) $e^{1/\pi}$ (B) $e^{-1/\pi}$ (C) $e^{-2/\pi}$ (D) $e^{2/\pi}$

58. $\int\limits_0^1\left(\sqrt[4]{1-x^7}-\sqrt[7]{1-x^4}\right)dx$ is equal to :

(A) 1/2 (B) 1 (C) 0 (D) None of these

59. Let $y'(x)+\frac{g'(x)}{g(x)}\cdot y(x)=\frac{g'(x)}{1+g^2(x)}$ where $f'(x)$ denotes $\frac{df(x)}{dx}$ and g(x) is a given non-constant differentiable function on R. If g(1) = y(1) = 1 and $g(e)=\sqrt{(2e-1)}$, then y(e) is equal to

(A) $\frac{3}{2g(e)}$ (B) $\frac{1}{2g(e)}$ (C) $\frac{2}{3g(e)}$ (D) $\frac{1}{3g(e)}$

60. If $z_1$ and $z_2$ satisfy $|z-(3+2i)|$ = Re z, and $\arg\left(\frac{z_1-4i}{z_2-4i}\right)=\frac{\pi}{2}$ then $\arg\left(\frac{z_1-(3+2i)}{z_2-(3+2i)}\right)$ is equal to:

(A) 0 (B) $\pi$ (C) $\pm\frac{\pi}{2}$ (D) None of these

61. The sum of the infinite series $\frac{1.3}{2}+\frac{3.5}{2^2}+\frac{5.7}{2^3}+\frac{7.9}{2^4}+\ldots\ldots\ldots\infty$ is :

(A) 32 (B) 23 (C) 29 (D) 30

62. Let A = {1, 2, 3, ……, 17}. For every nonempty subset B of A, the sum of the product of the reciprocals of the members of B is

(A) $\frac{153}{17!}$ (B) $\frac{153}{LCM(1,2,....,17)}$ (C) 18 (D) 17

63. If $S_n=\frac{\left(1^2+1-1\right)\cdot 1!}{1^2+3\cdot 1+2}+\frac{\left(2^2+2-1\right)\cdot 2!}{2^2+3\cdot 2+2}+\ldots..+\frac{\left(n^2+n-1\right)\cdot n!}{\left(n^2+3n+2\right)}$, then $S_6$ is equal to

(A) $\frac{1255}{2}$ (B) $\frac{1257}{2}$ (C) $\frac{1259}{2}$ (D) $\frac{1261}{2}$



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64. The largest value for x such that $\sum_{k=0}^{4}\frac{5^{4-k}}{(4-k)!}\left(\frac{x^k}{k!}\right)=\frac{8}{3}$ is

(A) $2\sqrt{2}+5$ (B) $2\sqrt{2}-5$ (C) $-2\sqrt{2}-5$ (D) $-2\sqrt{2}+5$

65. Suppose a solution of the differential equation $\left(x^7-x^3y\right)\frac{dx}{dy}=1$ satisfies the initial condition y(0) = 4, then the value of $\frac{dy}{dx}$ when $x=\sqrt{2}$ is

(A) $\frac{8}{e}$ (B) $8\sqrt{2}\left(1+\frac{2}{e}\right)$ (C) $8\left(1-\frac{2}{e}\right)$ (D) $8\sqrt{2}\left(1-\frac{2}{e}\right)$

66. Shortest distance between $|x| + |y| = 2$ and $x^2+y^2=16$ is

(A) $4+\sqrt{2}$ (B) $4-\sqrt{2}$ (C) 2 (D) 0

67. The coefficient of $x^{12}$ in the expansion of $\left(1+2x^2-x^3\right)^8$ is

(A) 6342 (B) 6343 (C) 6323 (D) 6349

68. If $\lim_{n\to\infty}\sum_{r=1}^{n}\cot^{-1}\left(r^2+\frac{3}{4}\right)=\tan^{-1}k$, then k is equal to

(A) 1 (B) 2 (C) 0 (D) None of these

69. Let the number $(22)^{2022}+(2022)^{22}$ leave the remainder $\alpha$ when divided by 3 and $\beta$ when divided by 7, then $\alpha^2+\beta^2$ is equal to

(A) 4 (B) 5 (C) 2 (D) 1

70. Let $f:R\to R$ and $g:R\to R$ be two functions defined by $f(x)=\log_e\left(x^2+1\right)-e^{-x}+1$ and $g(x)=\frac{1-2e^{2x}}{e^x}$, then the range of $\alpha$ for which the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right)>f\left(g\left(\alpha-\frac{5}{3}\right)\right)$ holds

(A) (2, 3) (B) (–2, –1) (C) (1, 2) (D) (–1, 1)



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## SECTION – II
## (Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

71. If the shortest distance between the lines $\vec{r_1} = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda\left(\hat{i} - 2\hat{j} + 2\hat{k}\right)$, $\lambda \in R$, $\alpha > 0$ and $\vec{r_2} = -4\hat{i} - \hat{k} + \mu\left(3\hat{i} - 2\hat{j} - 2\hat{k}\right), \mu \in R$ is 9, Then $\alpha$ is equal to

72. If $\alpha$ satisfies the equation $x^2 + x + 1 = 0$ and $(1+\alpha)^7 = A + B\alpha + C\alpha^2$, A, B, $C \geq 0$, then $5(3A - 2B - C)$ is equal to

73. Let x, y > 0, if $x^3y^2 = 2^{15}$, then the least value of $3x + 2y$ is equal to

74. If $\lim\limits_{x\to 0} \dfrac{\alpha x e^x - \beta \log_e(1+x) + \gamma x^2 e^{-x}}{x \sin^2 x} = 10$, where $\alpha, \beta, \gamma \in R$, then $\alpha + \beta + \gamma$ is equal to

75. Let $A = \{a_{ij}\}$ be a 3 × 3 matrix, where $a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j \\ 2 & \text{if } i = j \\ (-1)^{j+i} & \text{if } i > j \end{cases}$, then $\det\left(3\text{adj}\left(2A^{-1}\right)\right)$ is equal to



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